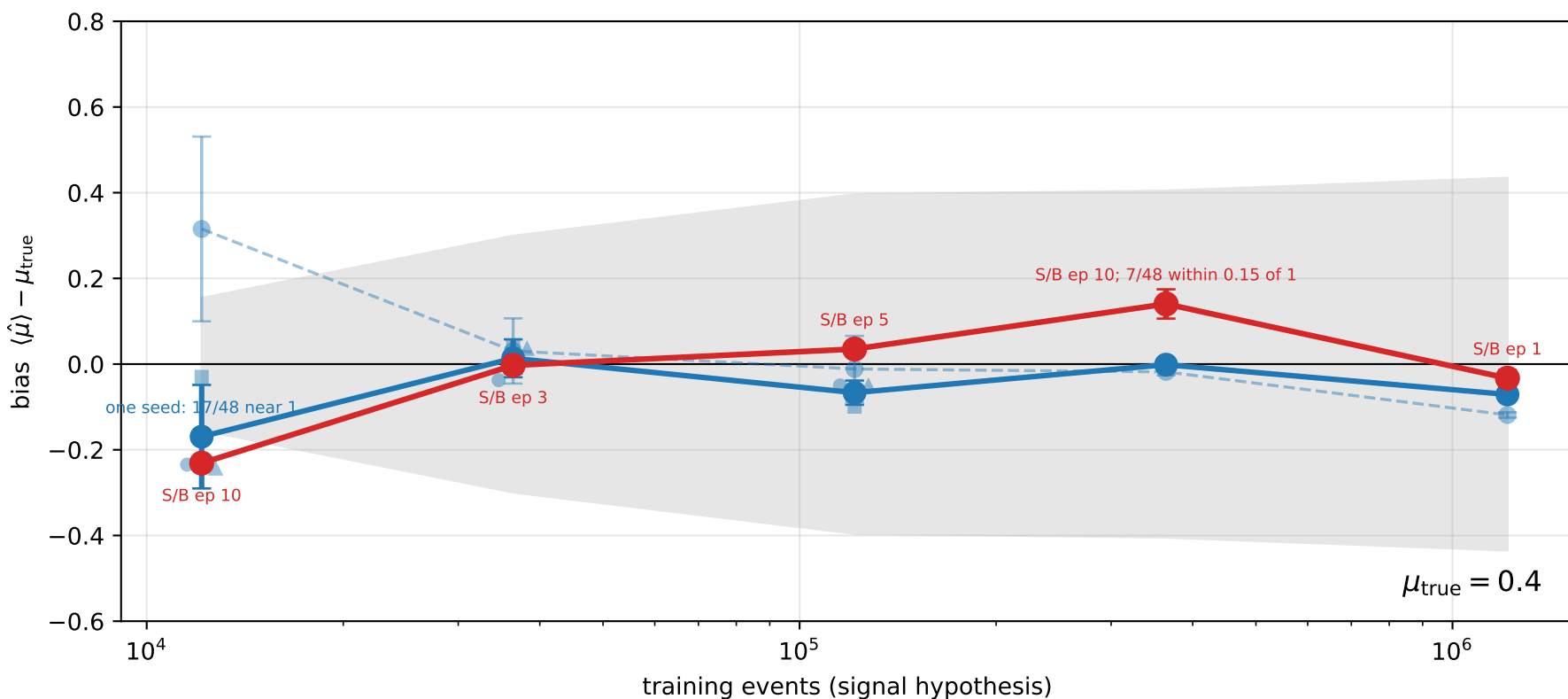
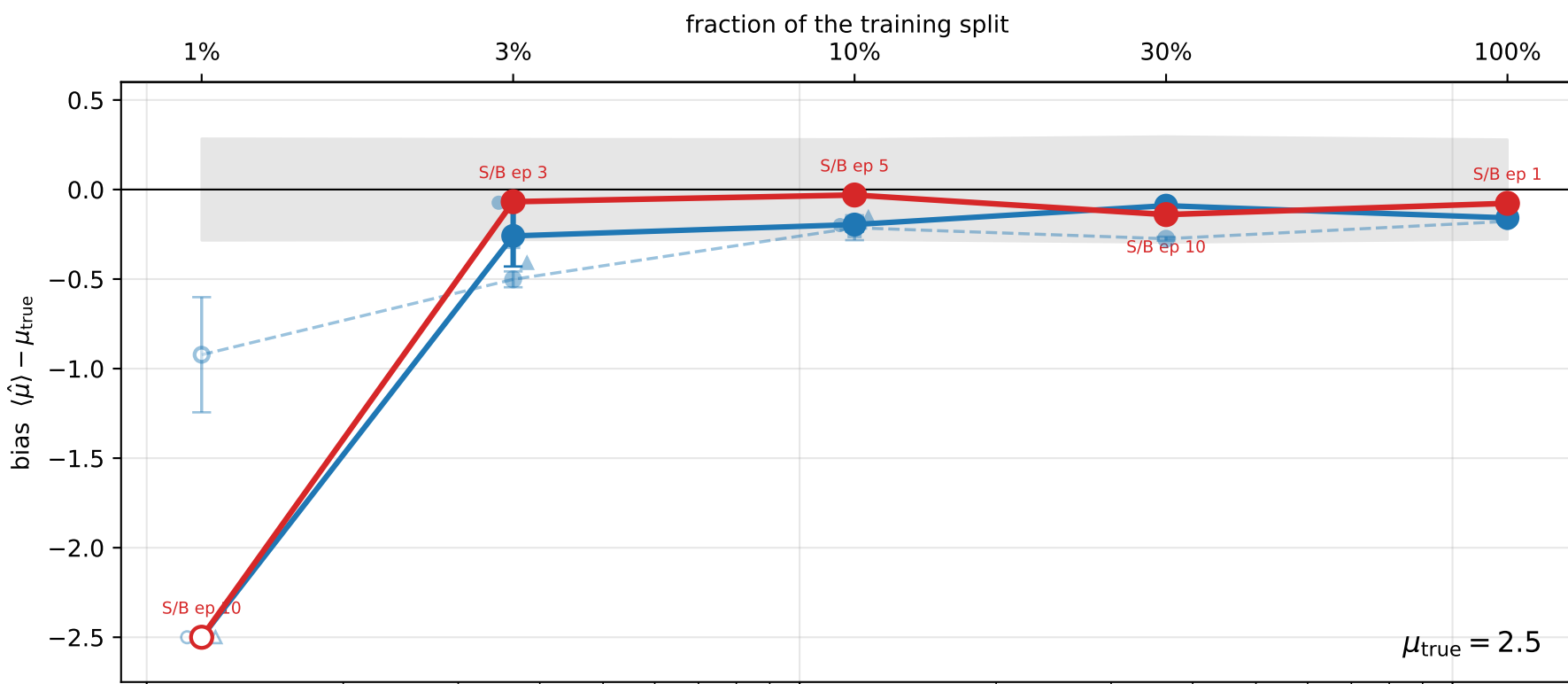


NSBI signal-strength bias vs training statistics: fine-tuned pretrained EveNet vs scratch
 best recipe of each model, both paired with the same fully trained SBI/B classifier; 48 weighted toys, 300 fb⁻¹



- pretrained, fine-tuned: S/B at the checkpoint with the smallest validation tail share at each size (no toys used) + SBI/B at epoch 20; seed 0
- scratch: S/B early-stopped best-val + the same SBI/B; mean over seeds 0/1/2 (small markers) at 1/3/10 %, seed 0 at 30/100 %
- reference: scratch exactly as run in the early-stopped scan (its own early-stopped SBI/B), mean over seeds
- $\pm\sigma_{\text{stat}}$: half of the mean 1 σ width of the scratch fits

error bars: seed spread where 3 seeds exist, otherwise toy s.e. only (excludes run-to-run spread, about 0.07 at 10 %).

Hollow: some 1 σ intervals open; points at -2.5: all 48 fits pinned at $\hat{\mu} = 0$.